

英文版 methods

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1. Experimental Procedure

1.1 Sample quality control

Please refer to QC report for methods of sample quality control.

1.2 Library preparation for Transcriptome sequencing

Non strand specific library

Messenger RNA was purified from total RNA using poly-T oligo-attached magnetic beads. After fragmentation, the first strand cDNA was synthesized using random hexamer primers followed by the second strand cDNA synthesis. The library was ready after end repair, A-tailing, adapter ligation, size selection, amplification, and purification. The library was checked with Qubit and real-time PCR for quantification and bioanalyzer for size distribution detection.

Strand specific library

Messenger RNA was purified from total RNA using poly-T oligo-attached magnetic beads. After fragmentation, the first strand cDNA was synthesized using random hexamer primers. Then the second strand cDNA was synthesized using dUTP, instead of dTTP. The directional library was ready after end repair, A-tailing, adapter ligation, size selection, USER enzyme digestion, amplification, and purification. The library was checked with Qubit and real-time PCR for quantification and bioanalyzer for size distribution detection.

1.3 Clustering and sequencing

After library quality control, different libraries were pooled based on the effective concentration and targeted data amount, then subjected to Illumina sequencing. The basic principle of sequencing is "Sequencing by Synthesis", where fluorescently labeled dNTPs, DNA polymerase, and adapter primers are added to the sequencing flow cell for amplification. As each sequencing cluster extends its complementary strand, the addition of each fluorescently labeled dNTP releases a corresponding fluorescence signal. The sequencer captures these fluorescence signals and converts them into sequencing peaks through computer software, thereby obtaining the sequence information of the target fragment.

2. Bioinformatics Analysis Pipeline

2.1 Data quality control

Raw data (raw reads) of fastq format were firstly processed through fastp software. In this step, clean data (clean reads) were obtained by removing reads containing adapter, reads containing ploy-N and low quality reads from raw data. At the same time, Q20, Q30 and GC content the clean data were calculated. All the downstream analyses were based on the clean data with high quality

The steps of data processing were as follows:

(1) Discard a paired reads if either one read contains adapter contamination;

- (2) Discard a paired reads if more than 10% of bases are uncertain in either one read;
- (3) Discard a paired reads if the proportion of low quality (Phred quality ≤ 5) bases is over 50% in either one read.

2.2 Reads mapping to the reference genome

Reference genome and gene model annotation files were downloaded from genome website. Use HISAT2 (2.2.1) to build the index of the reference genome, and use HISAT2 to align paired-end clean reads to the reference genome. HISAT2 can use the gene model annotation file to create splice-aware alignments, providing better alignment accuracy compared to other non-splice alignment tools.

2.3 Quantification of gene expression level

featureCounts (2.0.6) was used to count the reads numbers mapped to each gene. And then FPKM of each gene was calculated based on the length of the gene and reads count mapped to this gene. FPKM, expected number of Fragments Per Kilobase of transcript sequence per Millions base pairs sequenced, considers the effect of sequencing depth and gene length for the reads count at the same time, and is currently the most commonly used method for estimating gene expression levels.

2.4 Differential expression analysis

For DESeq2 with biological replicates: Differential expression analysis for two conditions/groups was performed using the DESeq2 R package

(1.42.0). DESeq2 provides statistical programs for determining differential expression in digital gene expression data using models based on negative binomial distribution. The resulting P-value is adjusted using the Benjamini and Hochberg' s methods to control the error discovery rate. The threshold of significant differential expression: $\text{padj} \leq 0.05 \ \& \ |\log_2(\text{foldchange})| \geq 1$.

For edgeR without biological replicates: Prior to differential gene expression analysis, for each sequencing library, read counts were adjusted using the edgeR R package (4.0.16) by scaling normalization factors to eliminate differences in sequencing depth between samples, followed by differential expression analysis. The resulting P value is adjusted using the Benjamini and Hochberg' s methods to control the error discovery rate. The threshold of significant differential expression: $\text{padj} \leq 0.05 \ \& \ |\log_2(\text{foldchange})| \geq 1$.

2.5 GO and KEGG enrichment analysis of differentially expressed genes

Gene Ontology (GO) enrichment analysis of differentially expressed genes was implemented by the clusterProfiler (4.8.1), in which gene length bias was corrected. GO terms with corrected P-value less than 0.05 were considered significantly enriched by differential expressed genes. KEGG is a database resource for understanding high-level functions and utilities of the biological system, such as the cell, the

organism and the ecosystem, from molecular-level information, especially large-scale molecular datasets generated by genome sequencing and other high-throughput experimental technologies (<http://www.genome.jp/kegg/>). We used clusterProfiler R package to test the statistical enrichment of differential expression genes in KEGG pathways.

2.6 Gene Set Enrichment Analysis

Gene Set Enrichment Analysis (GSEA) is a computational approach to determine if a predefined Gene Set can show a significant consistent difference between two biological states. The genes were ranked according to the degree of differential expression in the two samples, and then the predefined Gene Set were tested to see if they were enriched at the top or bottom of the list. Gene set enrichment analysis can include subtle expression changes. We use the local version of the GSEA analysis tool <http://www.broadinstitute.org/gsea/index.jsp>, GO, KEGG data set were used for GSEA independently.

2.7 PPI analysis of differentially expressed genes

PPI analysis of differentially expressed genes was based on the STRING database, which contains known and predicted protein-protein interactions. For species present in the database, we construct the network by extracting the target gene list from the database. Otherwise, we use diamond (version 0.9.13) to align the target gene sequences with

selected reference protein sequences, and then establish the network based on the known interactions of the selected reference species.

2.8 AS analysis

Alternative Splicing is an important mechanism for regulating gene expression and protein variability. The rMATS (4.3.0) software was used to analyze alternative splicing (AS) events, which mainly include five types of variable splicing events: SE (skipped exon), RI (retained intron), MXE (mutually exclusive exons), A5SS (5' alternative splice site), and A3SS (3' alternative splice site).

2.9 SNP analysis

The GATK (v4.1.1.0) software was used to perform SNP calling.

2.10 Fusion Analysis

Fusion gene refers to the chimeric gene formed by the fusion of all or part of the sequences of two genes, which is generally caused by chromosome translocation, deletion and other reasons. The STAR_fusion (version 1.12) software was utilized for the detection of fusion genes. SATR_fusion is a software package uses fusion output results of STAR alignment to detect fusion transcripts, including SATR alignment, SATRfusion.predict, SATRfusion.filter was used to correct the predicted results of SATR_fusion to ensure the accuracy of the results.

2.11 Weighted correlation network analysis

WGCNA (Weighted correlation network analysis) is a systematic biological method used to describe the gene association modes among different samples. it can be used to identify gene sets that are highly synergistic changed and identify candidate biomarkers or therapeutic targets based on the coherence of gene sets and the correlation between gene sets and phenotypes. WGCNA R package (1.71) is a set of functions used to calculate various weighted association analysis, which can be used for network construction, gene screening, gene cluster identification, topological feature calculation, data simulation and visualization. WGCNA is suitable for multisample data. Generally, more than 15 samples are required. One input file is sample information, that is, a matrix describing the traits of the sample: the traits used for association analysis must be numeric; If it is a regional or categorical variable, it needs to be converted to a 0-1 matrix. The other is gene expression data. For transcriptome sequencing, FPKM can be used as gene expression data.

2.12 Tumor immune microenvironment analysis

Tumor microenvironment (TME) is an important part of tumor tissue and plays a crucial role in tumor development. TME is composed of tumor cells, tumor stromal cells (including stromal fibroblasts), endothelial cells and immune cells (such as microglia, macrophages and lymphocytes), as

well as non cellular components of extracellular matrix (such as collagen, fibronectin, hyaluronic acid and laminin).

ESTIMATE is an R package that predicts tumor purity and uses gene expression data to predict the infiltration of stromal/immune cells in tumor tissue. It is based on the ssGSEA algorithm, and uses expression data to obtain stromal score and immune score in the test sample, to preliminarily evaluate the immune environment of individuals, subtypes, or cohorts. This analysis can not only obtain immune and matrix scores, but also comprehensively obtain the Estimate score, and calculate the tumor purity based on the Estimate score. The FPKM value of quantitative results is extracted as the input file, and ESTIMATE is used to calculate stromal cell scores, immune cell scores, and tumor purity. ggplot2 is used to conduct visualization displays (such as tumor purity bar charts and box plots).

xCell is an R package developed by the team that developed the SingleR package in 2017, which infers the proportion of cell types in population RNA samples. xCell comprehensively considers a total of 489 gene features from 64 cell types in six major databases. The ssGSEA algorithm is used to evaluate the scores of relevant gene features in each sample, and then calculate the scores of 64 cell types. This score considers the possibility of co-expression of the same gene between cells and evaluates the proportion of various cells in the sample. In the xCell

analysis, not only the scores of 64 types of cells can be obtained, but also the scores of immune cells, stromal cells and microenvironment cells can be obtained by comprehensive scoring. The FPKM value of the quantitative results is also extracted as the input file, and xCell is used to calculate the scores of immune cells, stromal cells and microenvironment, and calculate the percentage of different cell types in each sample. ggplot2 is used to conduct visualization displays such as histograms showing the proportion of scores for different cell types.

Immunophenogram (IPS) is specialized in evaluating immune related factors and predicting the response of immune checkpoint inhibitors to treatment. IPS divides 26 immune related factors into four subgroups: MHC molecule (MHC), immune checkpoint (CP), Effector cell (EC), and immunosuppressive cell (SC), which are four main types of genes (such as activated CD4+T cells, activated CD8+T cells, effector memory CD4+T cells, Treg, MDSCs, etc.). To evaluate the immune efficacy of this subgroup or immune factor in the body, IPS calculates the up and down expression of each immune factor by using the expression value of immune related genes. Meanwhile, IPS can comprehensively obtain an Immunophenoscore score to predict the treatment response of patients to immune checkpoint inhibitors. The FPKM value of the quantitative results is also extracted as the input file, and IPS is used to calculate the

immune related factor score. ggplot2 is used to conduct visualization displays such as immune related factor score representation diagram.